

## **U5-AT1 PROBLEM**

### **1. PROBLEM STATEMENT:**

Modern recommendation systems need to provide accurate and personalized suggestions to users while processing massive amounts of user-item interaction data. Traditional recommendation approaches may struggle with scalability, sparse user preferences, and the cold-start problem when used with large datasets. The problem is to design a hybrid AI-based recommendation engine that combines content-based filtering and collaborative filtering using PySpark RDDs to process large-scale data efficiently. The system should analyze user preferences, item characteristics, ratings, and interaction history in a distributed environment. It should also incorporate intelligent agents that continuously observe user behavior and feedback to adapt and improve recommendations in real time.

### **2. PROBLEM STATEMENT:**

Natural disasters such as floods and earthquakes can cause significant damage when they are not detected and responded to quickly. Disaster monitoring involves large and diverse amounts of information obtained from satellite images, environmental sensors, weather systems, and social media, making real-time analysis challenging. The problem is to develop a distributed AI-based disaster detection system using PySpark RDDs to process and analyze multi-source data efficiently. The system should combine information from different sources through data fusion techniques to identify possible disaster events, estimate their severity, and generate early warnings. Intelligent agents should coordinate the monitoring, analysis, warning, and response processes to support faster and more effective disaster management.

### **3. PROBLEM STATEMENT:**

The rapid growth of digital banking, online payments, and financial transactions has increased the risk of fraudulent activities. Financial institutions need to analyze millions of transactions and identify suspicious behavior quickly, but traditional centralized fraud detection systems may face scalability and performance limitations when processing large datasets. The problem is to design a scalable AI-based fraud detection framework using PySpark RDDs that can distribute transaction processing and apply anomaly detection techniques to identify unusual patterns. The system should analyze transaction characteristics, customer behavior, and historical activity to detect potential fraud. Intelligent agents should collaborate to monitor transactions, identify suspicious activities, generate alerts, and learn from feedback to continuously improve fraud detection accuracy.

### **4. PROBLEM STATEMENT:**

The rapid growth of digital banking, online payments, and financial transactions has increased the risk of fraudulent activities. Financial institutions need to analyze millions of transactions and identify suspicious behavior quickly, but traditional centralized fraud detection systems may face scalability and performance limitations when processing large datasets. The problem is to design a scalable AI-based fraud detection framework using PySpark RDDs that can distribute transaction processing and apply anomaly detection techniques to identify unusual patterns. The system should analyze transaction characteristics, customer behavior, and historical activity to detect potential fraud. Intelligent agents should collaborate to monitor transactions, identify suspicious activities, generate alerts, and learn from feedback to continuously improve fraud detection accuracy.

5. **PROBLEM STATEMENT:**

Educational institutions generate large amounts of student-related data, including learning activities, assessment results, attendance, interaction patterns, and course progress. Conventional learning systems often provide the same content to all students and may fail to identify individual learning difficulties or adapt to changing student needs. The problem is to design an AI-driven personalized learning analytics platform using PySpark RDDs that can efficiently process large-scale student data and identify individual learning patterns and performance trends. The system should use intelligent agents to continuously monitor student behavior, provide real-time feedback, recommend suitable learning materials, and adapt the learning path according to each student's progress. The platform should be scalable enough to support thousands of learners while improving personalization, engagement, and learning outcomes.